1. Exercises - I

(1) Let ω be a linear symplectic form on a vector space V. Consider the product vector space $V \times V$ with the form $\Omega = p_1^*\omega - p_2^*\omega$, where $p_1, p_2 : V \times V \to V$ are the projection morphisms onto the first and the second factor. Prove the following.

(a) Ω is a symplectic form.

(b) The diagonal subset $\Delta : \{(v, v) | v \in V\}$ is a Lagrangian subspace of $V \times V$.

(2) Let $Sp(2n)$ denote the group of all linear isomorphisms $A : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ which preserve the canonical symplectic form ω_0, that is, $A^*\omega_0 = \omega_0$. Show that, if λ is an eigen-value of A then λ^{-1} and $\bar{\lambda}$ are also its eigenvalues.